PHIRE · HENRY ICE RISE · DRAFT FOR COMMENT

How long, and how often, to measure borehole temperature

Survey design for recovering the undisturbed ice temperature after drilling, accurately enough to date the grounding of the ice rise

October 2026

Two design questions

QUESTION 1

One return visit, or two?

Is a reading a year after drilling close enough to the undisturbed ice temperature, or do we need to go back after two years?

QUESTION 2

Near-continuous logging, or a few readings?

Logging is possible with the right kit. The alternative is a few readings within weeks of installing the instrument, then one a year later.

The measurements are for dating when the ice rise grounded, so both questions come down to whether the temperatures are accurate enough for that. What we would like from you: whether the reasoning holds, what the models are missing, and which design you would choose.

The measurement problem

Drilling warms the ice

Making the hole holds its wall at 0 °C for somewhere between a day and a week. That heat spreads into the ice around it.

We can only measure in the hole

There are no sensors in the surrounding ice. The temperature in the borehole itself is the only record we get.

The warmth fades slowly

The borehole cools back towards the temperature the ice had before drilling, over years rather than weeks.

The design question is how late, and how often, we must read the borehole temperature to recover that undisturbed value.

Accuracy needed to date the grounding

The purpose of the measurements is to date when the ice rise grounded. A second model, of the temperature in a column of ice after it grounds, gives the sensor accuracy this needs.

60 mK	30 mK	20 mK	10 mK
grounded 1,000 years ago	grounded 2,000 years ago	grounded 3,000 years ago	grounded 5,000 years ago

Approximate sensor error that dates the grounding to within 10 %, when the geothermal flux, accumulation rate, surface temperature and velocity profile are fitted from the same profile. It is for 25 sensors whose errors are independent of each other.

Column model: 500 m of ice, −27 °C at the surface. These site values are placeholders, not measurements from Henry Ice Rise.

Model of the recovery

Heat conduction outward from the hole, in one dimension (radius), solved numerically.

Ice before drilling	−5 °C everywhere
Hole wall during drilling	held at 0 °C for 1 day or for 7 days
Hole wall afterwards	no heat flow; temperature free to relax
Hole radius	5 cm
Borehole reading	the temperature at the hole wall

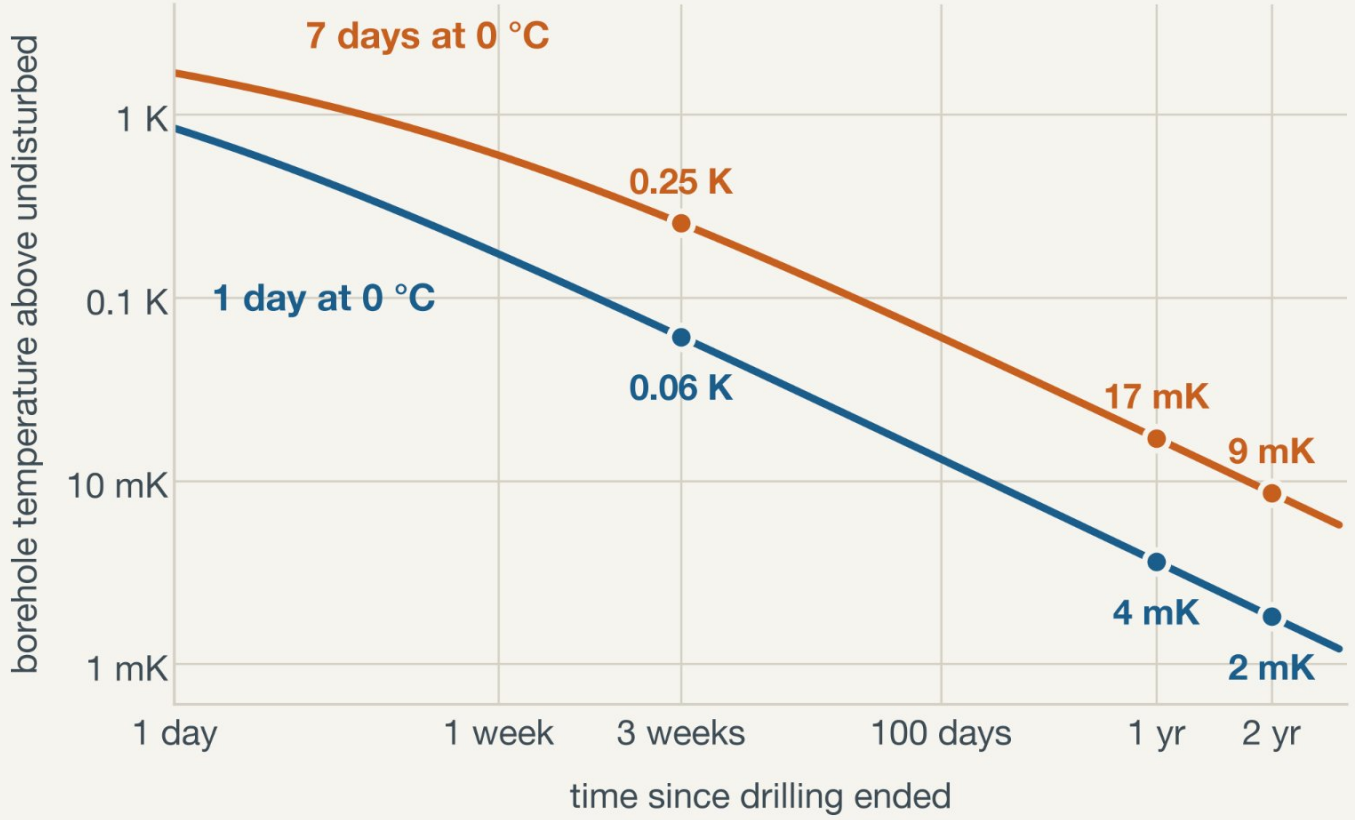
under 1 mK

difference from the exact solution for a cylinder held at a fixed temperature (Carslaw and Jaeger)

×5 at −25 °C

every error here scales with how far the ice is below 0 °C, so colder ice gives proportionally larger errors

Size of the disturbance



Borehole temperature above the undisturbed value. Each range runs from 1 day to 7 days with the wall at 0 °C.

0.06–0.25 K
three weeks after drilling

4–17 mK
one year after

2–9 mK
two years after

Model results for ice at −5 °C and a hole radius of 5 cm.

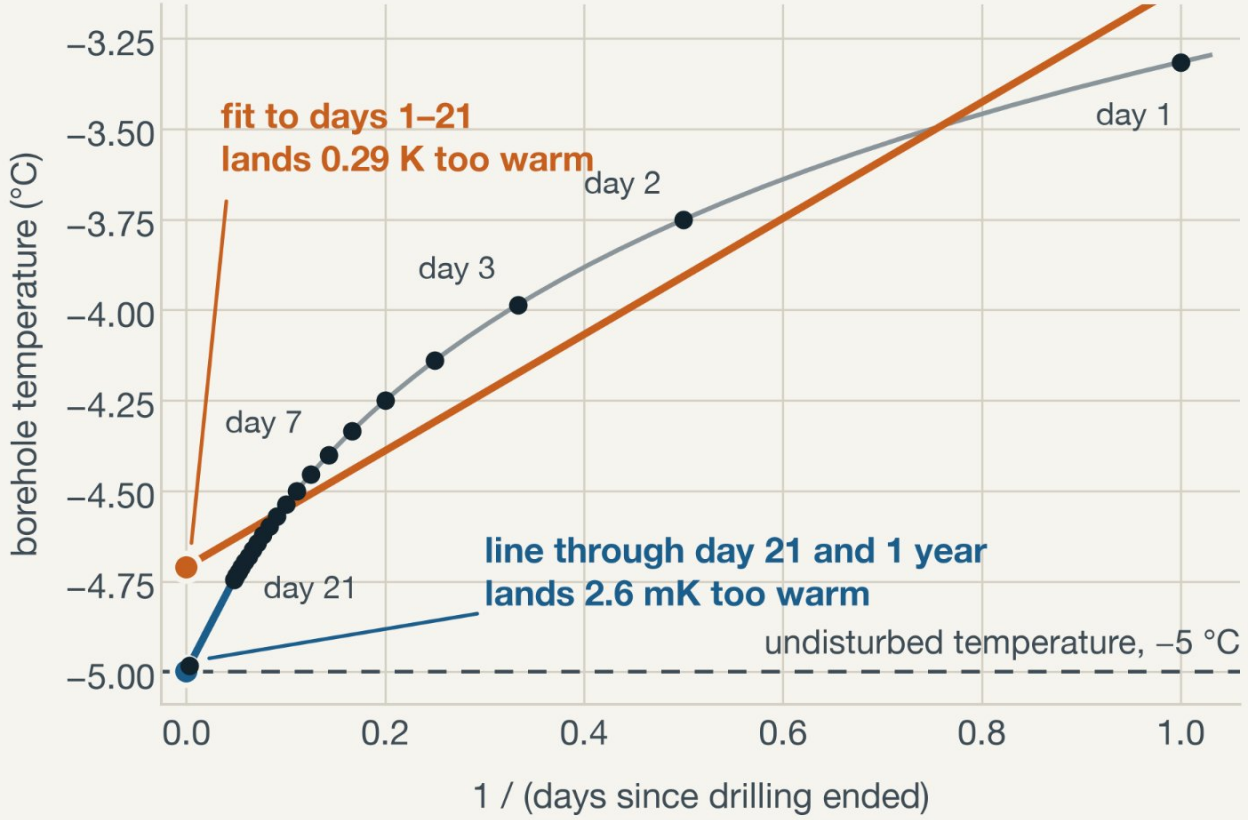
Extrapolation to the undisturbed temperature

Long after drilling, the borehole temperature T follows

$$T = T_0 + Q / (4\pi k\, t)$$

so a plot against 1/t is a straight line that points at the undisturbed temperature T0 (Humphrey and Echelmeyer, 1990).

This only holds once t is several times the heating duration s. A line fitted to readings taken around time t still leaves about s / (s + 2t) of the disturbance.



Example: 7 days at 0 °C, ice at −5 °C. Readings are model output with no measurement noise.

Question 1: one year or two

Error in the estimated undisturbed temperature, in millikelvin. In each cell the first figure is for 1 day at 0 °C and the second for 7 days.

What goes into the estimate	After the year-1 visit	Adding a year-2 visit
Latest reading, taken at face value	3.6 / 17	1.8 / 8.6
Year-1 and year-2 readings joined by a 1/t line (reading noise ×2.2)	needs both visits	below 0.1 / 0.1
Readings on days 7, 14 and 21 plus the visit readings, fitted with a free time shift	below 0.1 / 0.5	below 0.1 / 0.3
Daily logging, best choice of days, including 1 mK of reading noise	0.1 / 0.2 to 0.5	below 0.1 / 0.1 to 0.2

A second visit halves the error of a face-value reading. Once first-year readings are extrapolated, it adds little.

Model results for ice at −5 °C and a hole radius of 5 cm. The free time shift fits T = T0 + A / (t + t0).

Question 2: a few readings or a logger

Method error in millikelvin: 1 day / 7 days at 0 °C. Reading days are counted from the end of drilling.

Readings and fit	Method error (mK)	Reading noise passed on	Needs heating duration
One reading at 1 year, taken at face value	3.6 / 17	×1	no
Days 7, 14, 21 and 1 year: plain 1/t fit	2.3 / 36	×0.8	no
Day 21 and 1 year: 1/t line	0.1 / 2.6	×1.1	no
Day 21 and 1 year: ln(1 + s/t) line	below 0.1 / 0.3	×1.1	yes
Days 7, 14, 21 and 1 year: free time shift	below 0.1 / 0.5	×1.1	no
Daily logging for a year: best days and fit	below 0.1 / 0.1–0.4	×0.1 to ×0.3	no

With early readings only in week one, the 7-day errors grow to 5.7 mK (1/t line) and 2.6 mK (time shift). A twofold error in the heating duration s moves the ln line by about 1.5 mK.

Ice at −5 °C, hole radius 5 cm. Reading noise passed on: the share of one reading's random error that reaches the estimate.

Trade-offs between the two designs

A few readings, then one at a year

Method error of a few millikelvin or less, provided the last early reading comes about three heating durations after drilling and the early readings do not dominate the fit.

The single late reading passes its full noise, and any sensor drift, straight into the answer.

Nothing in the data shows whether the cooling followed the model.

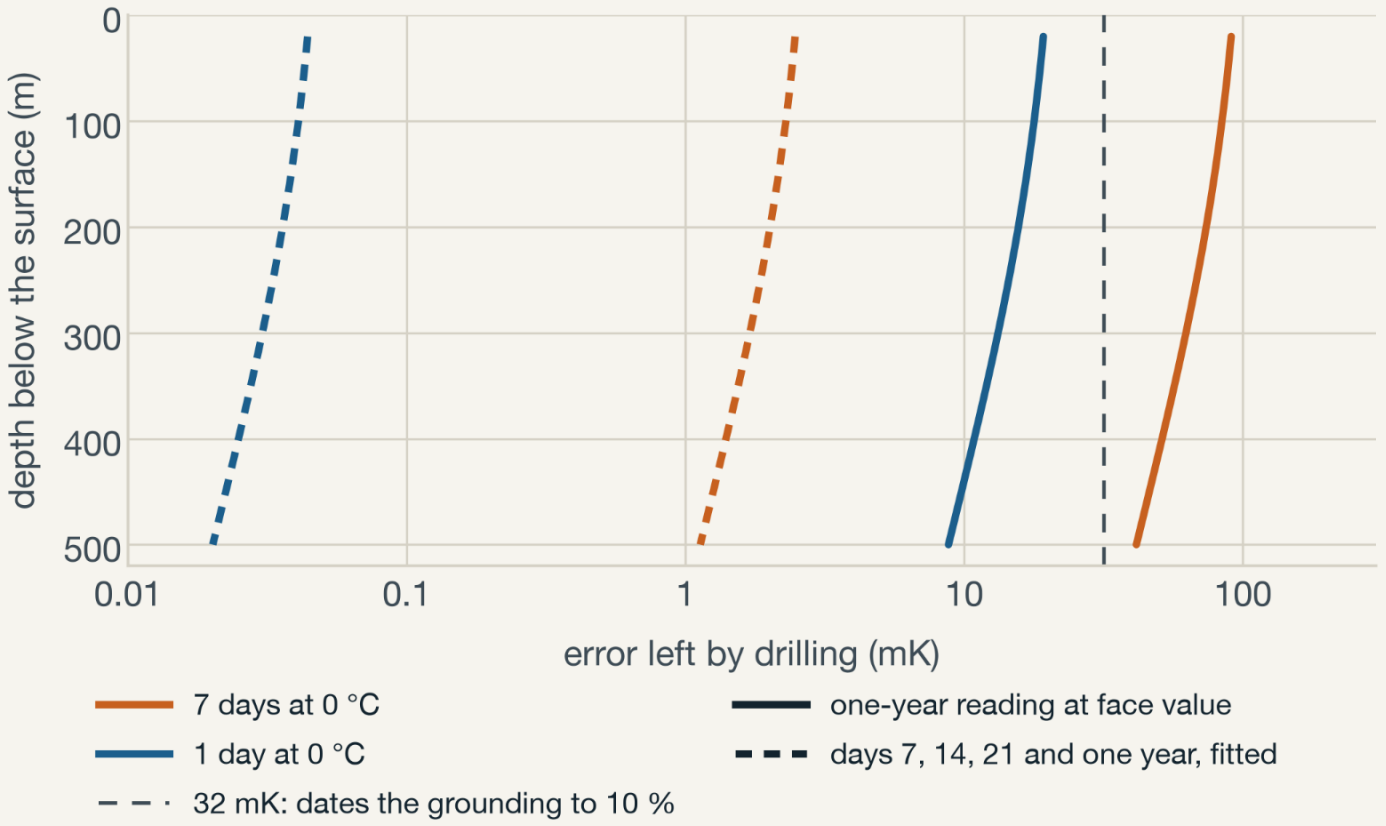
Near-continuous logging

Similar method error, with the freedom to choose afterwards which readings to fit.

Daily readings average random noise down about tenfold. Sensor drift is not averaged away.

The record shows the shape of the cooling, so the model and the time shift can be checked against it.

Heat left by drilling along the borehole



Error at each sensor one year after drilling. The ice is −27 °C near the surface and −12 °C at the bed.

40–90 mK

one-year reading at face value, after 7 days at 0 °C

9–20 mK

the same after 1 day at 0 °C

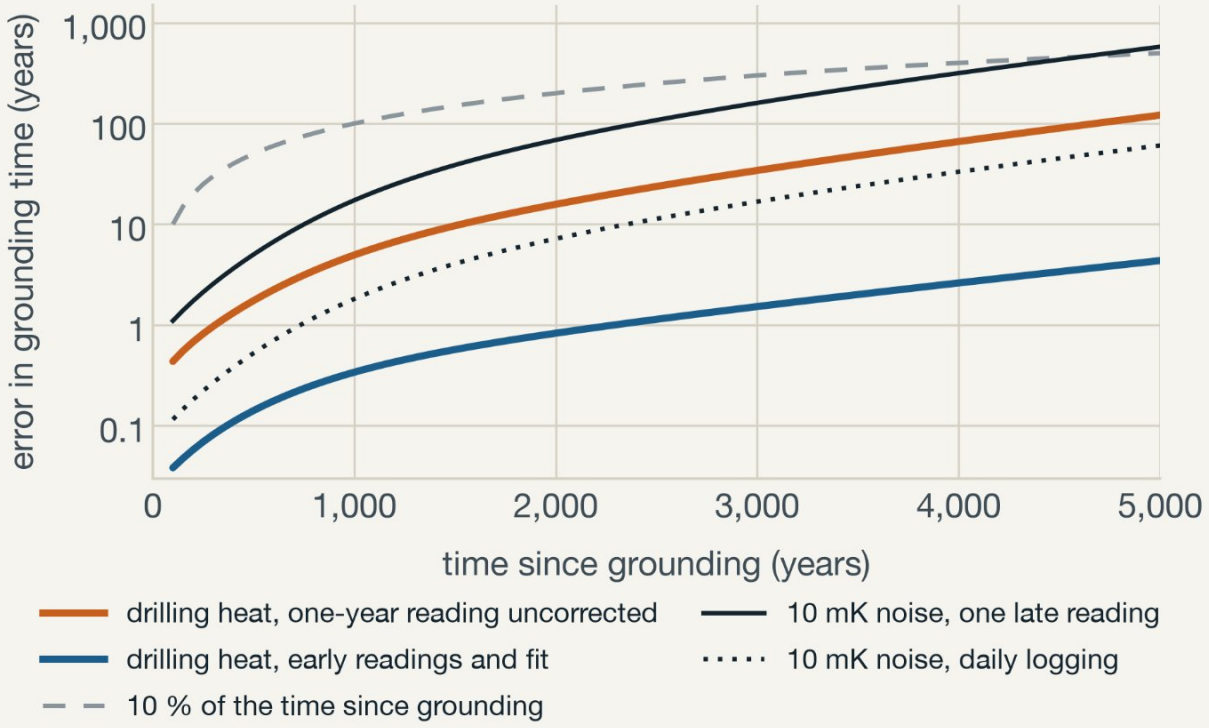
under 3 mK

with readings on days 7, 14 and 21 and the time-shift fit

Column 2,000 years after grounding. 32 mK is the independent sensor error that dates the grounding to 10 % at that age.

Effect on the grounding time

The drilling error is smooth and has the same sign at every sensor. The fit puts most of it into the surface temperature and the geothermal flux, so the grounding time moves little.



about 5 %

largest shift from an uncorrected one-year reading, in any case tried

under 1 %

with readings on days 7, 14 and 21 and the time-shift fit

17–580 years

from 10 mK of noise in one late reading; a tenth of that with a logger

Chart: heating for 1 day at the top to 7 days at the bed; other unknowns fitted. Noise range is for 1,000–5,000 years.

WHAT THE MODELS SAY

One return visit, a year after drilling, is enough. Readings in the first three weeks remove almost all the heat left by drilling. After that, sensor accuracy limits how well the grounding can be dated, not the schedule.

Limits of the models

Refreezing

No latent heat in the model. A water-filled hole stays at 0 °C until it refreezes, so the heating lasts longer than the drilling, varies with depth and is not well known.

Sensors

Reading noise is treated as random and independent. A constant offset, or a drift over the year, passes straight into the answer in every design.

Geometry

Heat flows outward only. There is no vertical temperature gradient, no ice flow, and no change in the true ice temperature during the survey.

Site values

The ice-rise column (500 m, −27 °C at the surface) uses placeholder values. The two question tables are for ice at −5 °C; errors scale with how far the ice is below 0 °C.

Questions for you

1. Is holding the hole wall at 0 °C for a set time a fair stand-in for drilling and refreezing? How long, and how would it vary with depth?
2. How stable should we expect the sensors to be over one to two years?
3. Is fitting a free time shift standard practice, and are there pitfalls we have missed?
4. Would you still plan a visit at two years, and why?

Sources

Borehole model, survey designs and grounding-time results: the notebook Radial heat diffusion around a borehole on the PHIRE site (github.com/jkingslake/PHIRE), sections 11 and 12.

Accuracy needed to date the grounding: the notebook Vertical temperature profile of an ice rise after it grounds on the same site, section 9.

Humphrey, N. and Echelmeyer, K. (1990). Hot-water drilling and bore-hole closure in cold ice. Journal of Glaciology 36(124), 287–298. The 1/t relation is used as quoted by McDowell et al. (2021), The Cryosphere 15, 897–907, supplement.

Lachenbruch, A. H. and Brewer, M. C. (1959). Dissipation of the temperature effect of drilling a well in Arctic Alaska. U.S. Geological Survey Bulletin 1083-C. Source of the ln(1 + s/t) form.

Carslaw, H. S. and Jaeger, J. C. (1959). Conduction of Heat in Solids, second edition. Exact solution used to check the model.